Exercise 1

1. Open a new excel document. Name the worksheet ***Apartment Managers***.

2. Format the worksheet as below:

	A	B	C	D	E
1	**Apartment Managers**				
2					
3	**Listing of Managed Apartments**				
4					
5	**Building**	**Street**	**Suburb**	**No of Apartments**	**Building Manager**
6	Green Park	324 St Kilda Road	Melbourne	76	Bill Smith
7	Sea View	324 Beaconsfield Pde	Albert Park	34	Bill Jones
8	Waves	324 The Esplanade	St Kilda	146	Bill Brown
9	Hollyaoks	324 Cantebury Rd	Middle Park	24	Bill Green
10	Lavender Grove	324 Park Street	South Yarra	19	Bill White
11	The Botanical	324 Domain Road	South Yarra	45	Bill Black
12	The Avenue	324 High Street	Prahran	165	Bill Blue
13	Gosford Park	324 Commercial Road	Prahran	110	Bill John
14					
15	No of Apartments				

3. Insert the formula in cell D15 using an appropriate function to find the total number of apartments managed.

4. Sort into lowest to highest order based on the number of apartments.

5. Copy the entire worksheet into a new worksheet (**Hint**: You can use the *Move or Copy* tool by right-clicking the worksheet on the worksheet tab) and rename it as ***ApManager_Sort***.

6. In ***ApManager_Sort*** worksheet, sort into A-Z order based on the suburb.

7. Format the cells from A17 onwards as the image below:

Building Manager Search

Building Managed	Building Manager
Sea View	

8. In cell C19, insert the formula using an appropriate `LOOKUP` function to find correct building manager based on the building stated in B19. (Hint: Specify the `range_lookup`). Apply absolute reference to the formula where necesssary.

Exercise 2

1. In the same workbook, open a new worksheet and rename it ***FruitsSales***.

2. Format the worksheet as below:

	A	B	C	D	E	F	G
1	Sales and Produce Department						
2	**Fruit Item**	**January**	**February**	**March**	**April**	**Total**	**Average**
3	Bananas	435	254	213	365		
4	Pears	345	482	550	216		
5	Grapes	279	469	347	523		
6	Oranges	461	227	681	560		
7	Apples	358	456	680	785		
8							
9		**Amount (RM)**	**Fruit Type**				
10	**Minimum**						
11	**Maximum**						
12							

3. Format the following cells:
   a. A1:G1 – Merge the cells, and fill with *Dark Blue, Text 2* color. Change the text to
      White and size 14pt.
   b. A2:G2 – Fill with *Tan, Background 2* color.
   c. A9:C9 – Fill with *White, Background 1, Darker 25%* color.

4. Under the Total and Average column, insert the formula using an appropriate function to find the total and average amount for each fruit.

5. In cell B10 and B11, insert the formula using an appropriate function to find the minimum and maximum total amount of fruits sold respectively.

6. In cell C10, insert the formula using an appropriate `LOOKUP` function to display the type of fruit which corresponds to the minimum total amount in cell B10 (**Hint**: `lookup_range` has to be sorted in ascending order). Apply the same formula in C11 to find the type of fruit which corresponds to the maximum total amount in B11. Apply absolute reference to the formula.

7. Formal all the currency values with a RM symbol and no decimal place.

8. Create a Clustered Column Chart from the month of January to April with the following formatting:
   a. Chart title: Sales and Product Department
   b. X-axis title: Month
   c. Y-axis title: Amount (RM)
   d. Place the legend at the bottom of the chart.

9. Create a 3-D Pie Chart to reflect the total amount of fruits sold. Remove the legend.

Insert an appropriate title for the chart and add the category name and value data labels to the chart.

10. Both charts are placed in the same worksheet below the data table.

Exercise 3

1. In the same excel document, create a new worksheet with the name ***Retail Product***. Format the worksheet as below:

Current date and time:

Product	Wholesale Price
Desk Chair	RM300.00
Blender	RM120.00
Coffee Maker	RM454.00
Laptop	RM2,300.00
Smartphone	RM1,650.00
Smart Watch	RM190.00
Sunglasses	RM59.00
Backpack	RM85.00
Dumbbell	RM32.00
Running Shoes	RM240.00

Mark-up % 82%

Profit and Demand of Retail Products For 4th Quarter

Product ID	Product Name	Supplier Name	Sales Quantity	Available Quantity	Wholesale Price per unit (RM)	Retail Price per unit (RM)	Profit per unit (RM)	Sales Profit (RM)	Reorder Point	To Reorder	Demand
101	Laptop	Tech and Co. Electronics	128	55					30		
102	Smartphone	Tech and Co. Electronics	207	132					50		
103	Running Shoes	ActiveStyle Emporium	150	170					20		
104	Blender	KitchenElite Sdn. Bhd.	83	15					15		
105	Backpack	UrbanCarry Essentials	90	90					10		
106	Desk Chair	Home Goods MY	34	25					17		
107	Coffee Maker	KitchenElite Sdn. Bhd.	103	22					25		
108	Sunglasses	UrbanCarry Essentials	115	95					15		
109	Dumbbell	ActiveStyle Emporium	60	44					45		
110	Smart Watch	Tech and Co. Electronics	166	72					40		

Total Profit

2. Insert the formula for the current date and time in cell C2. Apply "Thick Outside Borders" to the same cell.

3. The data for the products and their respective wholesale prices has been given to you in cells B5:C14. Name these cells ***ProdWS***.

4. Merge the cells A18:L18.

5. Format the fonts to the following cells:
   - A18:L18 – Calibri, 16 pt, bold and italics, fill colour "Blue, Accent 1, Lighter 40%", centre-aligned.
   - A19:L19 – Segoe UI, 9.5 pt, bold, fill colour "Blue, Accent 1, Lighter 60%", centre-aligned.
   - A20:A29 – Calibri, 11 pt, fill colour "Blue, Accent 1, Lighter 60%", centre-aligned.
   - The rest of the worksheet – Calibri, 11pt.

6. In cells F20, insert the formula using an appropriate `LOOKUP` function to find the wholesale price per unit for each product based on the pricing table in Step 3. Use ***ProdWS*** as `table_array` or `array`.

7. In cells G20, insert the formula to find the retail price per unit. The retail price is the wholesale price of the % in cell C16 and adds the wholesale price (Hint: that means if the mark-up percentage is 100% and the product's wholesale price is RM 2, the retail price will be RM 4). Apply absolute reference to the formula.

8. In cell H20, insert the formula to find the profit per unit. The profit per unit is the retail price minus the wholesale price.

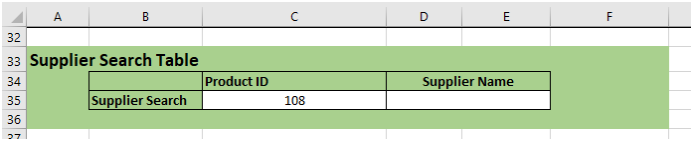
9. In cell I20, insert the formula or function to find the sales profit. The sales profit is profit per unit multiplied by the sales quantity.

10. Format all the currency values with a RM symbol and 2 decimal places.

11. In cell K20, use the `IF` function with the conditions below. **Centre-align your data:**
    - If the Available Quantity is less than or equal to the Reorder Point, then return the word "Reorder",
    - otherwise, return the word "No".

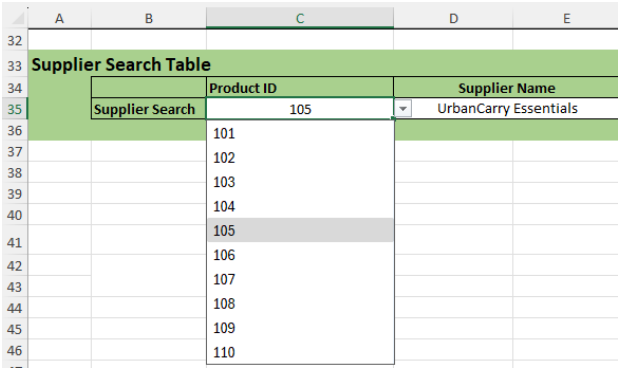
12. In cell L20, use the `IF` function with the conditions below. **Centre-align your data**:
    - If the Sales Quantity is more than 100 and the reorder point is more than 25, then return the word "HIGH",
    - else, if the Sales Quantity is more than 100 and the reorder point is less than or equal to 25, then return the word "AVERAGE",
    - otherwise, return the word "LOW".

13. Format A33:F36 as shown below:



    - Merge cells D34:E34 and D35:E35.
    - Fill colour – "Green, Accent 6, Lighter 40%", except for cells C35 and D35.
    - A33 – Calibri, 14 pt, bold.
    - B35, C34, D34 – bold.

14. In cell C35, insert data validation list to display the product ID from Column A.
15. In cell D35, use the `INDEX-MATCH` function to display the supplier name that corresponds to the Product ID in cell C35. The screenshot below shows the end product:



16. Find the Total Profit in cell I31 with the appropriate formula.

17. Save your work periodically to prevent data loss.